



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (Semester I) - August, 2024
STA1013 - Introduction to Statistics (3 Credits)

Answer ALL questions.

Time allowed: **TWO hours.**

Instructions

- This is a closed book examination.
- You are permitted to use a Scientific calculator during the examination.
- Use the answer booklets to write your answers and the single sheets for rough work.
- For any unacceptable behaviour, including the violation of examination regulations during the examination, official actions will be taken under Section 9 (Regulations Related to Exam Procedures) of the student handbook.

1. (a) Clearly define the following statistical terms. Include an example for each term to illustrate your definition. [6 marks]
- Discrete vs. Continuous
 - Population vs. Sample
 - Outlier
- (b) A distribution in which the mean, median and mode coincide is called as a symmetrical distribution.
- In moderately skewed or asymmetrical distribution, write down the relationship among the three measures of central tendency. [4 marks]
 - For a moderately skewed data, the arithmetic mean is 200, the coefficient of variation is 8 and Karl Pearson's coefficient of skewness is 0.3. Calculate the mode and the median of the dataset. [5 marks]
- (c) For a dataset containing 10 observations, mean and sample standard deviations are 20 and 6, respectively. It was found after the calculations were done that observations 5 and 15 are incorrect. Find the mean and the standard deviation if these two incorrect observations are omitted from the dataset. Show all your work. [10 marks]
2. (a) Derive the expressions for the third and fourth central moments, μ_3 and μ_4 , in terms of the first four central moments about the origin. [6 marks]
- (b) Given $\beta_1 = 1.0$, $\beta_2 = 4.0$ and Variance = 9,
- Find the values of μ_3 and μ_4 . Here μ_i is the i^{th} moment about the mean. [6 marks]
 - Describe the nature of the distribution of the data. [3 marks]
- (c) (i) Explain any advantages and disadvantages of using a stem-and-leaf plot compared to other graphical representations like histograms or boxplots. [3 marks]
- ii. Given the following weights (in grams) of 20 packages:
- | | | | | | | | | | |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 221 | 235 | 197 | 184 | 210 | 152 | 275 | 263 | 143 | 192 |
| 276 | 215 | 183 | 178 | 260 | 195 | 200 | 140 | 174 | 155 |
- Construct a stem-and-leaf plot for this dataset. [4 marks]
- iii. Based on the shape of the distribution of the stem-and-leaf plot you obtained in part (ii), determine an approximate % of data that falls within two standard deviations about the mean. [3 marks]

[P.T.O.]

3. (a) An agronomist experimented to observe the relationship between the amount of fertilizer used and the yield. For this experiment, he selected eight tomato plants of the same variety and weekly treated with x grams of fertilizers to randomly selected plants. Subsequently, he recorded the yield (y) in kilograms.

Plant	A	B	C	D	E	F	G	H
Fertilizer (x) grams	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5
Yield (y) kilograms	3.9	4.4	5.8	6.6	7.0	7.1	7.3	7.7

$n = 8$

- Plot a scatter diagram of yield (y) against the amount of fertilizer (x). [2 marks]
- Interpret your observation using the above plot. [2 marks]
- Calculate the correlation coefficient between the amount of fertilizer (x) applied and the yield (y). $x \quad y \quad x^2 \quad y^2 \quad xy$ [6 marks]
- Interpret the result obtained in (iii). [2 marks]

- (b) i. Show that the least square estimates of the coefficients of the simple linear regression are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \text{ and } \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad [4 \text{ marks}]$$

- ii. For the data given in part (a), find the coefficients of the simple linear regression of y on x. [8 marks]

- iii. Estimate the yield of a plant treated weekly with 3.2 grams of fertilizer. [1 mark]

4. (a) State the equations for calculating price index under the following indexing methods.

- Simple Relative Index. [1 mark]
- Simple Aggregate Index. [1 mark]
- Simple Average of Relatives Index. [1 mark]

- (b) Explain what are the weaknesses of Simple Relative Index, Simple Aggregate Index, and Simple Average of Relatives Index. [3 marks]

- (c) Following table shows the prices and consumption of three commodities in 1995 and 2005.

Commodity	1995		2005	
	Price	Quantity	Price	Quantity
A	5	60	15	70
B	4	20	8	35
C	3	15	6	20

- Calculate the following price index numbers for 2005 by
 - Laspeyres's method. [3 marks]
 - Paasche's method. [3 marks]
 - Fisher's method. [3 marks]
- Interpret the above results. [2 marks]
- Which price index number from Laspeyres's, Paasche's, and Fisher's would you recommended to use? [1 mark]
- Why did you make the above recommendation? [2 marks]

- (d) According to the Central Bank Report, average salary of a state sector executive-level officer was Rs. 22,935/= in 2015. It is claimed that a same-level employee is currently (2024) earning Rs. 47,615/=. Also, the Consumer Price Index (CPI) was reported as 108.2 in 2015 January and 200.7 in 2024 January.

- Calculate the real wage of the above employee. [3 marks]
- Interpret your answer. [2 marks]